

Nitrogen Metabolism of Early Lactation Cows Fed Diets with Two Different Levels of Protein and Different Amino Acid Profiles¹

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ABSTRACT

Four multiparous Holstein cows (569 ± 122 kg) surgically prepared with indwelling catheters in the mesenteric, portal, and hepatic veins and carotid artery were allocated in a 4×4 Latin square to determine the effects of dietary crude protein (CP) level and amino acid (AA) profile on N metabolism during early lactation (from 25 to 65 d in milk). Cows received their diets in two equal meals and were milked twice daily. The dietary treatments were: 18% CP with a high (18H) or a low (18L) quality AA profile, and 15% CP with a high (15H) or a low (15L) quality AA profile. The four diets were similar in net energy for lactation ($1.75 \text{ NE}_L \text{ Mcal/kg}$) and contained the same amount of RUP (34% of CP). The quality of the AA profile pertained only to the essential AA (EAA), and was assessed by comparison with the EAA profile of casein and considered the potential contribution of EAA from ruminal bacteria. The 18H and 15H diets were supplemented with 50 and 25 g/d of ruminally protected Met, respectively. After 10 d on treatment, a blood flow marker (p-amino-hippurate) was infused into a mesenteric vein, and arterial, portal, hepatic, and mammary blood samples were obtained at 3, 6, and 12 h after feeding. Dry matter intake was similar across treatments (23.4 ± 0.5 kg/d). Amino acid oxidation, and consequent urea production, in the liver were numerically greater with the 18% CP rations, and, as a result, arterial urea concentrations were greatest ($P < 0.01$) with these rations. The amount of total AA extracted by the mammary gland tended to be greater with the H than with the L diets (21.4 vs. 18.2 mmol/h, respectively). Milk yield tended to be greater ($P =$

0.16) with the 18H and 15H diets (47.7 and 46.3 kg/d, respectively) compared with the 18L and 15L diets (45.9 and 44.6 kg/d, respectively). Also, milk CP and casein contents were greatest ($P = 0.09$) with the H diets compared with the L diets. Milk and plasma urea N were greatest ($P < 0.01$) with the 18% CP diets. The efficiency of N utilization for milk protein synthesis was greatest ($P < 0.09$) with the 15% CP diets. It is concluded that milk protein production during early lactation is less susceptible to variations in dietary CP contents than variations in the AA profile of the dietary protein.

(Key words: amino acids, milk production, plasma flow, protein)

Abbreviation key: 18H = diet with 18% CP and a high quality AA profile; 18L = diet with 18% CP and a low quality AA profile; 15H = diet with 15% CP and a high quality AA profile; 15L = diet with 15% CP and a low quality AA profile, AV = arteriovenous, EAA = essential AA, MUN = milk urea nitrogen, PAH = para-aminohippurate, PDV = portal-drained viscera.

INTRODUCTION

There has been an increased interest in formulating diets and protein supplements for ruminants to provide a specific array of AA to the small intestine. One of the theoretical advantages of formulating rations with an adequate AA profile is that the amount of CP included in the rations can potentially be reduced, resulting in a positive economic impact. Metcalf et al. (31) fed three different levels of dietary CP (11, 15, and 20%) and found no differences in milk yield and milk protein yield between the rations with 15 or 20% CP. Similarly, Cunningham et al. (10) found no differences in milk yield or milk protein yield when feeding rations containing 16.5 or 18.5% CP to dairy cows in early lactation. In spite of these observations, current feeding practices recommend rations that contain at least 18% CP for early-lactation cows. Schwab et al. (37) emphasized that optimizing intestinal AA balance is more important to improving milk protein production than

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is the CP content of the ration. A potential consequence of an unbalanced AA profile flowing to the duodenum is that an excess of certain AA may reduce the possibilities of absorption of other AA, because AA are absorbed across the intestinal wall by an active transport mechanism (38). For example, Johns and Bergen (21) reported that an excess of Leu depressed the absorption of Lys *in vitro*, a phenomenon previously observed by Hume et al. (15). Ruminally protected individual AA, instead of whole protein sources, offer the potential to balance rations for meeting AA needs of the animal without resulting in an excess of undesirable AA, which could interfere with intestinal absorption due to competition for the same AA transporter. The objective of this study was to determine the effects of dietary CP level and AA profile on nitrogen metabolism and milk production of dairy cows during early lactation.

MATERIALS AND METHODS

Animals and Treatments

Ten weeks before calving, four multiparous dairy cows were surgically prepared with indwelling catheters in the hepatic, portal, and two mesenteric veins (17). The surgeries and the experiment were conducted under the approval and supervision of the Animal Care Committee of the University of Minnesota. The study began when the animals reached 25 DIM. At that time, the cows were allocated in a 4 × 4 Latin square design. The cows were confined to tie stalls with free access to water. The average BW of the animals at the beginning of the experiment was 569 ± 122 kg. Each experimental period was 10 d long. To avoid potential problems with catheter patency and based on the similarity of the ingredients used with the different dietary treatments, we felt that 10-d periods were adequate. The four dietary treatments were 18% CP with a high (18H) or a low (18L) quality AA profile, and 15% CP with a high (15H) or a low (15L) quality AA profile. The diets were formulated using linear programming with Mathematica (Wolfram Research, Champaign, IL; 43). The quality of the AA profile referred only to the essential AA (EAA), and was assessed by comparison with the EAA profile of casein and accounted for the potential contribution of EAA from ruminal bacteria. The bacterial contribution to AA flow to the duodenum was estimated based on the relationship between OM intake and microbial flow described by Clark et al. (8).

Once the microbial flow was estimated, the difference between the amount of AA provided by bacteria and those estimated to be needed to produce 45 kg of milk was used as a constraint in the linear programming model. Because of the lack of sufficient information, it was assumed that the AA profile of the protein re-

maining after ruminal incubation would be similar to the AA profile of the protein before ruminal incubation. One of the constraints used in the model was RUP that was set to 35% of CP, with the 65% remaining available to maximize microbial protein production in the rumen. Also, maximum concentrations of each EAA to achieve an optimum EAA profile were set for the 18H and 15H diets to resemble the AA profile of casein.

To achieve high dietary concentrations of Met without increasing the concentrations of other AA, 50 and 25 g/d of a ruminally protected Met source (Met-Plus, Nisso America, New York, NY) were included in the 18H and 15H diets, respectively. Similarly, these two diets were supplemented with 20 and 10 g/d of an unprotected form of L-Lys. The inclusion of unprotected Lys had the purpose of supplying additional Lys to the intestine. Lysine is extensively degraded in the rumen; however, the degradation of Lys is time dependent, as it is for any other AA, peptide, or protein. A large proportion of the supplied Lys would be degraded, but a small fraction could escape the rumen. The 15L and 18L diets were balanced, forcing the use of similar ingredients used in the formulation of 18H and 15H but constraints on EAA profile were removed. The ingredient and chemical composition and the AA profiles of the diets are depicted in Tables 1 and 2, respectively. Cows received their daily diets at 0400 and 1600 h in two equal meals and were milked at 0600 and 1800 h. Feed refusals were collected and weighed each day before the p.m. feeding.

Sample Collection and Chemical Analyses

On the last day of each experimental period, blood samples (25 ml) were withdrawn simultaneously into heparinized syringes from the hepatic and portal veins and the carotid artery at 3, 6, and 12 h after the a.m. feeding. An additional blood sample was taken from the carotid artery and prepared as serum, by allowing coagulation and then centrifuging at 1500 × *g* at 4°C for 15 min. Immediately after these blood samples were collected, 25 ml of blood from the caudal epigastric vein was obtained by venipuncture. Blood samples were transferred into tubes containing 20 IU of heparin and 1.5 mg of dried NaF and centrifuged at 1500 × *g* at 4°C for 15 min to obtain plasma. Serum samples were used for insulin and glucose analyses.

Serum insulin was determined by radioimmunoassay (Coat-a-Count Insulin, Diagnostic Products Corporation, Los Angeles, CA), and serum glucose was determined colorimetrically (kit number 510; Sigma Chemical Co, St. Louis, MO). Plasma samples were used to determine ammonia, urea, NEFA, and free AA content. Analytical determinations of ammonia, urea, and free AA content were performed on a 6300 Amino Acid Ana-

Table 1. Ingredient and chemical composition of the experimental diets.

	Dietary treatment ¹			
	18H	18L	15H	15L
- Ingredient composition, % of DM -				
Ingredient				
Alfalfa hay	11.77	19.60	10.86	16.50
Corn silage	28.48	28.45	34.12	37.32
Corn grain	17.44	17.42	17.34	17.36
Corn distillers with solubles	6.49	...	6.52	...
Soybean hulls	4.35	5.67	6.52	6.52
Cottonseed with lint	14.34	7.41	14.29	6.95
Soybean meal	12.12	8.72	5.22	1.30
Corn gluten meal	0.44	0.66	0.39	0.52
Soybean seed	...	7.84	...	8.68
Animal fat	0.44	0.87	0.87	1.30
Met-Plus ²	0.22	...	0.11	...
L-Lysine	0.08	...	0.04	...
Mineral-vitamin mix	3.28	3.38	3.67	3.59
— Chemical composition ³ —				
Item				
Crude protein	18.3	17.8	14.9	14.7
RUP ⁴ , % of CP	35.9	32.7	36.5	32.4
NE _L , Mcal/kg ⁵	1.76	1.74	1.76	1.72
Neutral detergent fiber	31.2	30.0	33.5	31.5
Acid detergent fiber	20.2	20.1	21.3	20.5
Nonfibrous carbohydrates ⁵	35.5	36.3	36.1	38.0
Lipid	6.3	6.5	6.8	6.8

¹18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

²Met-Plus is a matrix of DL-methionine (65%), calcium salts of long chain fatty acids, a fatty acid and talc commercialized by Nisso-America, Inc., NY.

³Values are expressed as a % of DM unless otherwise indicated.

⁴RUP: Rumen undegradable protein.

⁵All values represent actual chemical measurements of the diets, with the exception of NE_L and nonfibrous carbohydrates which were estimated mathematically.

lyzer (Beckman Instruments, Fullerton, CA) after the addition of (S)-2-aminoethyl-1-cysteine as an internal standard and precipitation of plasma proteins with sulfosalicylic acid (15% wt/vol). Plasma NEFA concentrations were determined with a colorimetric method (NEFA-C kit, Wako Chemical Industries USA, Inc., Richmond, VA). Plasma flow through the portal-drained viscera (PDV) was measured by the dilution technique described by Katz and Bergman (23) using para-aminohippurate (PAH). At least 20 min before and during each sampling time, a sterile solution containing 10% PAH (wt/vol at a pH of 7.4) was constantly infused with a screw-driven syringe constant-infusion pump (Harvard Apparatus Co., Holliston, MA) at a rate of 2.7 ml/min through a mesenteric vein, after the infusion of a priming dose of 38 ml of the same solution. The concentration of PAH in plasma was determined

with an HPLC (model 603, Beckman Instruments, Fullerton, CA) fitted with a Hypersil C18 reversed-phase column (5- μ m particle size, 250- $\times$ 4.6-mm i.d.; Alltech Associates, Inc., Deerfield, IL) following the procedure described by Bell et al. (3).

Grab samples of each diet were collected throughout each experimental period and composited at the end by treatment. Feed samples were ground through a 2-mm screen, and then analyzed for CP with the macro-Kjeldahl procedure (1), for neutral and acid detergent fiber following the sequential analyses described by Van Soest et al. (42), for RUP with the in situ technique and for lipid content as ether extract (1). Amino acid content was determined on a Beckman 6300 Amino Acid Analyzer (Beckman Instruments, Palo Alto, CA) using norleucine as an internal standard to correct for AA recovery.

Milk yield was recorded at the a.m. and p.m. milking times during the last 5 d of each period. Milk samples were collected at the same times and analyzed for total N and casein content following the procedures described by the International Dairy Federation (18, 19). Milk fat and milk urea nitrogen (MUN) were measured by infrared analysis (Minnesota DHIA, Zumbrota, MN).

Calculations and Statistical Analyses

Plasma flow rates and net metabolite fluxes across the PDV and liver were calculated using the Fick principle as described elsewhere (2). The Fick principle and the assumption that the sum of milk Phe and Tyr outputs can be equated with the sum of their mammary

Table 2. Amino acid profile (% of total AA) of the experimental diets.

Amino acid	Dietary treatment ^a			
	18H	18L	15H	15L
Alanine	7.13	6.57	6.41	6.22
Arginine	3.21	1.04	4.15	4.02
Aspartate	11.59	10.96	9.66	10.68
Cysteine	2.29	2.32	2.62	3.08
Glutamate	12.21	19.68	17.20	16.02
Glycine	4.83	5.23	4.50	4.89
Histidine	2.76	2.45	2.81	2.59
Isoleucine	5.00	4.87	4.29	4.43
Leucine	9.72	9.15	10.24	9.44
Lysine	5.13	5.29	4.77	5.13
Methionine	2.38	1.54	2.04	1.70
Phenylalanine	5.18	5.70	5.13	5.51
Proline	6.34	7.11	6.56	6.96
Serine	5.31	5.01	4.91	5.12
Threonine	4.62	4.35	4.02	4.10
Tyrosine	5.68	2.31	4.31	3.59
Valine	6.63	6.42	6.39	6.52

^a18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

Table 3. Effects of dietary CP and the AA profile offered to dairy cows in early lactation on plasma flows (L/h) in the portal, hepatic and mammary vein.¹

Vein	Dietary treatment ²				SE	Probability ³		
	18H	18L	15H	15L		AA	CP	CP × AA
Portal	1608	1760	1639	1754	104	0.33	0.89	0.81
Hepatic	2054	2143	2053	2187	121	0.43	0.81	0.80
Mammary	937	909	900	909	138	0.86	0.45	0.53

¹Results are expressed as least square means with $n = 48$ except for the mammary plasma flow which has $n = 16$.

²18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

³Probability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: Crude protein level in the ration (18 or 15).

arteriovenous plasma extractions (11) were used to estimate plasma flow through the mammary gland as follows: milk phenylalanine + tyrosine outputs / arteriovenous difference. Phenylalanine and Tyr concentration in milk protein were assumed to be 5.1 and 4.9 g/100 g of milk protein, respectively (20). For these calculations, it was also assumed that 86% of milk protein was synthesized in the mammary gland (22).

Data regarding metabolite flux through the PDV, arterial metabolite concentrations, and milk yield and composition were analyzed as a repeated measures design with the MIXED procedure of SAS (36) to fit the following model:

$$\begin{aligned} \text{dependent variable} = & \mu + \text{cow} + \text{period} + \text{AA} + \text{CP} \\ & + \text{AA} \times \text{CP} + \text{Cow} \times \text{AA} \times \text{CP} + \text{Time} \\ & + \text{AA} \times \text{Time} + \text{CP} \times \text{Time} + \varepsilon; \end{aligned}$$

where μ is the grand mean and ε is the random error. Time referred to hours after feeding for the data regarding plasma metabolites, and to collection day for the data related to milk production. The error term used for testing treatment effects was defined by the interaction $\text{Cow} \times \text{AA} \times \text{CP}$.

The data regarding the mammary plasma flow (Table 3) and mammary metabolite flux (Table 8) were analyzed as a Latin square without repeated measures, combining the data from each collection time into an average for each treatment, cow, and period. These data were fitted using the GLM procedure of SAS (35) using the following model:

$$\begin{aligned} \text{dependent variable} = & \mu + \text{cow} + \text{period} + \text{AA} + \text{CP} \\ & + \text{AA} \times \text{CP} + \varepsilon \end{aligned}$$

where μ is the grand mean and ε is the random error.

For all the analyses, the significance level was set to 0.10, with the values ranging from 0.10 to 0.20 being considered trends.

RESULTS AND DISCUSSION

Plasma Flows and Dry Matter Intake

The approach used to estimate mammary plasma flow in the current study has also been used by others (11), and it has been shown to result in lower variation compared with the dye dilution technique (5). The ratio between blood flow and milk yield averaged about 650, which is greater than previous studies that reported approximately 575 L of blood are required to produce 1 kg of milk (29). Plasma flows through the portal, hepatic, and mammary veins were not affected by treatments (Table 3). Similarly, DMI was the same among treatments, with an average of 23.4 kg/d (Table 4). Plasma flows through the portal and hepatic veins were greater than those reported previously (34, 40) with dairy cattle in early lactation. A possible explanation for the higher flows observed in the current study could be the greater DMI and consequently greater energy intake, compared with intakes reported in previous studies.

Milk Production Responses

Milk yield was not affected by the level of CP of the dietary treatments (Table 4), although milk yield tended to increase ($P = 0.16$) with the diets containing a high quality AA profile (H) compared with those containing a low quality AA profile (L). Cows that received the 15L diet produced 3.1 kg/d less milk than the cows that received the 18H ration; however, cows receiving the 15H diet produced only 1.4 kg/d less milk than the cows receiving the 18H diet.

Within each level of protein consumption, a high quality AA profile tended to increase milk protein and casein yields. The H rations resulted in greater milk CP percentages ($P = 0.07$) and CP yield ($P < 0.09$), and milk casein content ($P = 0.09$) and casein yield ($P = 0.10$) than the L rations. This increase in milk protein

and casein contents, as a response to a high quality AA profile, tended to be more accentuated ($P = 0.16$) with the 15% than the 18% CP rations, suggesting that the AA profile appears to gain importance as the amount of CP in the ration diminishes. In contrast to the response obtained with a high quality AA profile, the amount of dietary CP appeared to have little effect on milk CP ($P = 0.20$) or casein ($P = 0.93$) contents. However, the amount of dietary CP significantly affected MUN concentrations. The 18% CP rations had greater ($P < 0.01$) MUN concentrations than the 15% CP rations, which paralleled arterial urea concentrations. This observation agrees with previous literature (6) in which positive correlations between dietary CP and MUN concentrations were reported.

The efficiency of dietary AA-N utilization for milk AA-N synthesis was highest ($P < 0.09$) with the 15% CP rations. The efficiencies were 33, 30, 27, and 26% for the 15H, 15L, 18H, and 18L rations, respectively. Therefore, it is concluded that during early lactation, reductions in dietary CP concentrations without a depression in milk production are possible, thereby resulting in an increased efficiency of N utilization for milk protein synthesis.

Milk fat content tended to increase ($P = 0.16$) with the H rations compared with the L rations. This increase was greater with the 18% CP rations compared with the 15% CP rations, which is in contrast with the observations on milk protein content. This fact might also be related to the numerically greater energy intake observed with the 18 compared with the 15% CP rations. Milk fat yield tended to increase ($P = 0.11$) with the rations containing a high quality AA profile; this increase was larger for the 18% than the 15% CP ra-

tions. Methionine supplementation has been reported to increase milk fat content and yield (32); however, the mechanisms by which this increase in milk fat occurs is unknown, although it could be related to an increase in choline synthesis from Met (39) or an improvement in the synthesis of plasma lipoproteins (27).

Portal Metabolite Flux

The net portal plasma appearance of ammonia N ranged from 34 to 37% of the total N absorbed by the PDV and was about 19% of total N intake, whereas the net appearance of AA-N in plasma across the PDV ranged from 31 to 37% of total N intake. Similar observations have been reported elsewhere (2, 44). Total N appearance across the PDV in the form of AA plus ammonia ranged from 49 to 56% of N intake. There was a net flux of urea from plasma toward the lumen of the gut, which ranged from 21 to 23% of N intake. Urea N transfer from plasma to the PDV has been reported to range from 10 to 42% of N intake (16). Urea N flux from plasma to the PDV tended to be greater with the 18 (average 22.3% of N intake) than with the 15% CP rations (average 21.6% of N intake), which was probably due to the greater arterial urea concentrations obtained with the 18 compared with the 15% CP rations.

Overall, portal AA-N flux was 17.8 and 9.9% greater than the milk AA-N output with the 18H and 18L diets, respectively. However, the portal AA-N flux was 12.7 and 10.9% lower than the milk AA-N output with the 15H and 15L rations, respectively. Alanine was the AA with the greatest net portal fluxes (Table 5) followed by Gly, Leu, and Ser, which is similar to observations in a previous study (2). Net portal fluxes of those AA

Table 4. Effects of dietary CP and the AA profile offered to dairy cows in early lactation on dry matter intake, milk yield and milk composition.

Item	Dietary treatment ¹				SE	Probability ²		
	18H	18L	15H	15L		AA	CP	CP × AA
Dry matter intake, kg/d	23.5	23.7	22.9	23.3	0.50	0.50	0.33	0.86
Yield, kg/d								
Milk	47.7	45.9	46.3	44.6	1.05	0.16	0.25	0.97
Crude protein	1.60	1.50	1.53	1.39	0.06	0.09	0.18	0.73
Casein	1.28	1.19	1.25	1.12	0.06	0.10	0.41	0.67
Fat	1.86	1.71	1.69	1.62	0.08	0.11	0.15	0.56
Milk composition								
Crude protein, %	3.30	3.27	3.31	3.10	0.06	0.07	0.20	0.16
Casein, %	2.62	2.60	2.72	2.49	0.06	0.09	0.93	0.14
Urea nitrogen, mg/dl	19.09	17.66	12.48	11.33	0.86	0.19	<.01	0.88
Fat, %	3.88	3.72	3.68	3.66	0.07	0.16	0.26	0.24

¹18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

²Probability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: Crude protein level in the ration (18 or 15).

Table 5. Effects of dietary CP and the AA profile offered to dairy cows in early lactation on net portal plasma fluxes (mmol/h) of some metabolites.

Metabolite ³	Dietary treatment ¹				SE	Probability ²		
	18H	18L	15H	15L		AA	CP	CP × AA
Ammonia	396.3	385.4	302.6	293.9	54.26	0.86	0.14	0.99
Urea	-235.1	-216.4	-183.2	-168.1	68.12	0.81	0.48	0.98
Alanine	68.9	74.0	45.6	45.2	12.90	0.83	0.09	0.84
Arginine	32.0	26.5	22.6	19.1	3.14	0.20	0.04	0.77
Asparagine	28.2	31.8	23.1	19.1	3.88	0.96	0.06	0.37
Aspartate	3.7	2.1	1.8	2.0	0.65	0.30	0.16	0.23
Citruline	19.7	18.2	11.8	6.9	5.46	0.57	0.12	0.76
Cysteine	0.51	0.43	0.15	0.02	0.25	0.57	0.04	0.90
Glycine	45.4	43.2	44.5	39.4	9.33	0.71	0.81	0.88
Histidine	12.5	9.9	7.5	7.5	1.23	0.33	0.03	0.34
Isoleucine	34.5	30.9	23.9	22.5	4.73	0.62	0.09	0.83
Leucine	53.9	44.1	41.6	44.5	7.83	0.68	0.48	0.45
Lysine	33.5	34.7	31.7	27.1	3.45	0.64	0.22	0.43
Methionine	15.9	10.5	9.9	7.8	1.25	0.02	0.01	0.25
Ornithine	6.0	9.1	5.4	6.6	1.59	0.24	0.37	0.58
Phenylalanine	24.7	23.6	19.1	23.2	2.51	0.58	0.26	0.36
Proline	18.5	20.4	16.7	13.2	2.92	0.78	0.17	0.39
Serine	39.4	37.8	31.2	25.9	3.99	0.41	0.05	0.66
Taurine	-1.5	-7.2	-6.0	-8.6	4.43	0.39	0.53	0.74
Threonine	27.3	28.1	22.4	22.2	4.63	0.95	0.31	0.92
Tryptophan	5.2	4.3	3.1	5.3	2.19	0.76	0.82	0.50
Tyrosine	27.2	20.6	17.3	21.5	4.07	0.81	0.33	0.22
Valine	41.3	32.8	23.6	25.8	7.38	0.69	0.15	0.51

¹18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

²Probability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: CP level in the ration (18 or 15).

³Positive numbers indicate net release into plasma, whereas negative numbers indicate net removal from plasma.

with highest concentrations in microbial protein (Asp, Lys, and Leu) were not significantly different among treatments; however, net portal fluxes of those AA with the lowest concentrations in microbial protein (Cys, His, and Met) were affected ($P < 0.05$) by the CP level of the rations. Also, net portal fluxes of Arg and Ser (two AA that are also found in relatively low concentrations in microbial protein) were lower ($P < 0.05$) with the 15 than with 18% CP rations. These observations suggest that microbial growth in the rumen was probably similar with all the rations, and that the differences in net portal AA flux associated with the level of dietary CP might have been a consequence of a greater amount of dietary protein leaving the rumen with the 18% CP rations.

Net portal Met flux was affected ($P < 0.05$) by the AA profile of the rations. Because a source of ruminally protected Met was included in the H rations, a higher net portal Met flux was expected with the H rations in comparison with the L rations. This observation confirms that the ruminally protected form of Met used in the current study was successfully protected against ruminal degradation and well absorbed in the small intestine.

Hepatic and Splanchnic Metabolite Flux

Hepatic removal of plasma ammonia N accounted for 56, 63, 60, and 56% of the total urea N produced by the liver with 18H, 18L, 15H, and 15L, respectively. These values are within the range of 46 to 89% reported in the literature (26). Hepatic rates of fractional extraction of each plasma AA did not differ ($P > 0.30$) across treatments (data not shown). The most gluconeogenic AA (Glycine, Ala, and Ser) were removed in greatest quantities by the liver (Table 6), which is consistent with other studies (2, 44). In contrast, Phe, Tyr, and Met were the EAA with highest fractional hepatic extractions (9.3, 8.9, and 7.8% of hepatic supply, respectively).

The liver removal of most AA accounted for much of their net release from the PDV, with the exemption of the branched-chain AA (Ile, Leu, and Val) and Lys, which were removed by the liver in small amounts, and as a consequence these AA had the greatest net splanchnic fluxes (Table 7). Similar observations have been reported elsewhere (33). Net splanchnic flux of most AA was positive, with the exception of Ala, Gly, Ser, and Tau. Alanine, Gly, and Ser are the principal gluconeogenic AA and have been reported by Reynolds

et al. (33) to have negative net splanchnic fluxes, which indicate a net mobilization from body protein reserves. Total net splanchnic flux of AA-N, and thus the AA-N supply to peripheral tissues, tended to be greater ($P = 0.17$) with the 18H and 18L rations (135 and 125 g/d, respectively) than with the 15H and 15L rations (105 and 87 g/d, respectively).

The amounts of each AA relative to the amounts of Lys (depicted within parenthesis in Table 7) leaving the splanchnic tissues were very similar among treatments, which suggests that the liver removed any AA unbalances. Despite the different net portal Met fluxes among treatments, the hepatic fractional extraction of Met was similar among treatments, which resulted in a greater fraction of Met escaping liver metabolism and thus a greater ($P < 0.05$) net Met splanchnic flux with the 18 than with the 15% CP rations. Similarly, the net Met splanchnic flux was greater ($P < 0.10$) with the H rations than with the L rations, which could be due to the greater net portal Met flux consequence of feeding ruminally protected Met.

Mammary Metabolite Flux

The total amounts of AA being removed by the mammary gland across treatments (Table 8) were not differ-

ent. Plasma arteriovenous (AV) concentration differences observed in this study were similar to those reported by Drackley and Schingoethe (12) with cows in early lactation. Most of the EAA were removed by the mammary gland in excess of their output in milk protein, and several NEAA were removed by the mammary gland in quantities that were insufficient to account for their output in milk as observed in previous studies (30). Guinard and Rulquin (13) and Trottier (41) reported linear relationships between arterial concentrations of EAA and AV differences in the mammary gland. This relationship was also observed in the current study ($r = 0.56$; $P < 0.03$). However, there was a poor relationship ($r = 0$; $P = 0.87$) between AV differences of EAA and milk protein yield. To account for the correlation between arterial concentrations and AV differences, mammary uptake can be expressed as a ratio between mammary uptake and arterial supply, as proposed by Clark et al. (9). This approach could be an accurate method for predicting limiting AA for milk protein synthesis, because it is expected that the extraction of the limiting AA be much greater than the other AA. Using this approach, the limiting EAA for milk protein synthesis among all treatments appeared to be Lys, which was removed by the mammary gland at 68% of its arterial

Table 6. Effects of dietary crude protein level and the amino acid profile offered to dairy cows in early lactation on net hepatic flux (mmol/h) of some metabolites.

Metabolite ³	Dietary treatment ¹				SE	Probability ²		
	18H	18L	15H	15L		AA	CP	CP × AA
Ammonia	-421.4	-430.3	-356.5	-312.1	50.11	0.73	0.11	0.61
Urea	377.1	341.9	297.2	278.2	66.17	0.69	0.31	0.90
Alanine	-79.3	-88.0	-69.4	-57.2	16.80	0.92	0.27	0.56
Arginine	-12.5	-9.2	-6.2	-10.0	5.04	0.96	0.60	0.51
Asparagine	-17.6	-16.2	-12.0	-12.8	5.43	0.96	0.43	0.85
Aspartate	-1.5	-1.0	-0.92	-1.0	0.57	0.75	0.60	0.61
Citrulline	-10.7	-9.8	-3.1	0.05	8.22	0.81	0.33	0.89
Cysteine	.06	0.05	0.02	0.16	0.20	0.74	0.87	0.74
Glycine	-87.3	-77.5	-10.1	-10.2	1.68	0.81	0.30	0.75
Histidine	-8.1	-3.9	-5.4	-5.0	2.09	0.32	0.72	0.40
Isoleucine	-2.5	-5.8	-3.0	-6.2	2.30	0.20	0.84	0.99
Leucine	-4.9	-5.4	-6.5	-11.5	2.78	0.36	0.22	0.45
Lysine	-8.3	-5.6	-7.4	-10.8	3.38	0.93	0.54	0.39
Methionine	-5.9	-3.1	-3.6	-3.1	1.97	0.44	0.60	0.59
Ornithine	1.8	6.1	5.9	1.8	2.88	0.97	0.96	0.20
Phenylalanine	-9.5	-10.3	-7.7	-12.9	3.48	0.43	0.93	0.54
Proline	-12.1	-13.1	-9.1	-8.4	4.38	0.97	0.41	0.85
Serine	-42.0	-39.1	-40.8	-28.0	6.56	0.28	0.38	0.50
Taurine	-7.4	-3.5	-5.4	1.3	4.03	0.23	0.42	0.73
Threonine	-12.0	-15.7	-15.7	-16.4	4.20	0.67	0.68	0.78
Tryptophan	-3.0	-2.4	-2.5	-5.2	2.51	0.73	0.70	0.52
Tyrosine	-11.1	-7.6	-7.6	-10.9	4.06	0.91	0.91	0.39
Valine	-7.9	-9.4	-6.8	-8.1	3.86	0.73	0.73	0.98

¹18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

²Probability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: Crude protein level in the ration (18 or 15).

³Positive numbers indicate net release into plasma, whereas negative numbers indicate net removal from plasma.

Table 7. Effects of dietary crude protein level and the amino acid profile offered to dairy cows in early lactation on net splanchnic plasma fluxes (mmol/h) of some metabolites.¹

Metabolite ⁴	Dietary treatment ²				SE	Probability ³		
	18H	18L	15H	15L		AA	CP	CP × AA
Ammonia	-25.1	-44.9	-54.0	-18.2	38.58	0.84	0.98	0.50
Urea	141.9	125.5	114.0	110.1	46.26	0.83	0.66	0.90
Alanine	-10.4	-14.0	-23.9	-12.0	11.54	0.73	0.64	0.53
Arginine	19.4 (.77)	17.2 (.60)	16.4 (.68)	9.0 (.55)	4.61	0.34	0.27	0.60
Asparagine	10.5 (.42)	15.5 (.53)	11.0 (.45)	6.2 (.38)	4.17	0.98	0.33	0.29
Aspartate	2.2 (.09)	1.01 (.04)	.89 (.04)	0.92 (.06)	0.61	0.40	0.27	0.38
Citrulline	9.0 (.36)	8.4 (.29)	8.7 (.36)	7.0 (.43)	5.11	0.82	0.86	0.92
Cysteine	0.57 (.02)	0.48 (.02)	.17 (.01)	0.18 (.01)	0.19	0.85	0.13	0.81
Glycine	-41.9	-34.4	-56.5	-62.8	15.79	0.97	0.22	0.68
Histidine	4.5 (.18)	6.0 (.20)	2.1 (.09)	2.5 (.15)	1.82	0.63	0.16	0.75
Isoleucine	32.0 (1.27)	25.1 (.86)	20.9 (.86)	16.3 (1.00)	4.84	0.28	0.09	0.82
Leucine	49.0 (1.95)	38.8 (1.33)	35.2 (1.45)	33.2 (2.03)	7.78	0.45	0.25	0.62
Lysine	25.1 (1.00)	29.1 (1.00)	24.3 (1.00)	16.3 (1.00)	5.36	0.71	0.25	0.31
Methionine	10.0 (.40)	7.4 (.26)	6.3 (.26)	4.7 (.29)	1.12	0.10	0.03	0.67
Ornithine	7.8 (.31)	15.2 (.52)	11.2 (.46)	8.4 (.52)	2.88	0.47	0.58	0.13
Phenylalanine	15.2 (.60)	13.3 (.46)	11.4 (.47)	10.1 (.62)	2.56	0.57	0.22	0.93
Proline	6.4 (.26)	7.2 (.25)	7.7 (.32)	4.9 (.30)	3.63	0.79	0.88	0.63
Serine	-2.8	-1.3	-9.5	-2.2	6.16	0.49	0.56	0.64
Taurine	-8.9	-10.7	-11.4	-7.3	5.30	0.83	0.93	0.59
Threonine	15.0 (.60)	12.4 (.43)	6.8 (.28)	5.9 (.36)	4.20	0.68	0.13	0.83
Tryptophan	1.9 (.08)	1.9 (.06)	0.56 (.02)	0.20 (.01)	0.24	0.93	0.55	0.95
Tyrosine	15.3 (.61)	13.0 (.45)	9.7 (.40)	10.9 (.67)	4.83	0.92	0.45	0.72
Valine	33.4 (1.33)	23.4 (.80)	16.8 (.69)	17.7 (1.08)	6.33	0.52	0.14	0.44

¹Numbers within parenthesis represent the abundance of each amino acid leaving the splanchnic tissues relative to Lys.

²18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

³Probability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: Crude protein level in the ration (18 or 15).

⁴Positive numbers indicate net release into plasma, whereas negative numbers indicate net removal from plasma.

supply, immediately followed by Met (extracted at 61% of its arterial supply). Therefore, it is concluded that Lys and Met (in that order) were the limiting AA for milk protein synthesis in the rations used in the current study. The next limiting AA appeared to be Arg (extracted at 53% of its arterial supply), Phe (removed at 47% of its arterial supply), and Ile and Leu (extracted at 44 and 42% of their arterial supply, respectively). The least-limiting EAA appeared to be Trp, which was extracted by the mammary gland at a rate of 20% of its arterial supply.

Arginine was the EAA with the lowest efficiency of utilization in the mammary gland. The mammary gland removed Arg in amounts 2.6 times greater than those secreted in milk protein. Arginine has been consistently reported as an EAA that is removed by the mammary gland in much greater amounts than it is secreted in milk protein (4, 13, 31). A possible explanation for this observation is the participation of Arg in the synthesis of Pro, which is poorly extracted from plasma by the mammary gland (7). The EAA used with the greatest efficiency for milk protein synthesis were Met and Phe,

which were removed in amounts that were only slightly greater than their milk outputs (3 and 7% greater, respectively). Lysine, which according to its rate of mammary extraction seemed to be the limiting AA across treatments, was used by the mammary gland with lower efficiency than Met or Phe, although the efficiency of Lys utilization by the mammary was greater with the 15% CP rations (the mammary uptake was 26% greater than the milk output) than the 18% CP rations (mammary uptake was 37% greater than the milk output). Lysine was the AA that the mammary gland removed in the most consistent amounts across treatments, with a coefficient of variation across treatments of 5%, which would further support the theory that Lys was the first-limiting AA for milk protein synthesis across treatments. In contrast, the removal of Met by the mammary gland was 5% lower than the Met milk output with the 18% CP rations, whereas it was greater than the milk Met output with the 15% CP rations. This observation might imply that the mammary gland may have preferred to use plasma peptides as a source of Met for milk protein synthesis with the 18% CP ra-

tions. With the 18% CP rations, the mammary gland removed 58% of the arterial Met supply, whereas with the 15% CP rations the mammary gland removed 65% of the arterial Met supply.

The net splanchnic flux of AA-N was 95, 96, 104, and 91 g/d lower than the mammary AA-N uptake with 18H, 18L, 15H, and 15L, respectively, which is in agreement with the study of Reynolds et al. (34) who reported that the calculated net total splanchnic contribution of N to the peripheral tissues was 127 g/d less than measured output of milk and urine N. The difference between splanchnic supply and mammary uptake of AA could have been explained by some AA supply by the erythrocytes. Hanigan et al. (14) indicated that there is a poor correlation between mammary uptake of individual AA from plasma and from blood and that uptakes derived using plasma could not adequately represent whole blood AA uptake. However, in the current study only plasma AA were measured due to technical difficulties. Also, the contribution of peptide-bound AA, which may represent a potential source of some AA for the mammary gland (4) was not measured in this study either.

Arterial Insulin and Metabolite Concentrations

Arterial insulin concentrations were greater ($P < 0.05$) with the 18 than with the 15% CP rations (Table

9). Several authors demonstrated that an increase in supply of AA to the animal may result in greater release of insulin (24, 25). In spite of differences in arterial insulin concentrations, arterial glucose concentrations did not differ among treatments (Table 9), although they were numerically greater with the 15H ration. Also, the greater ($P < 0.05$) arterial insulin concentrations observed with the 18% CP rations coincided with a numerical increase in milk fat concentration (Table 4), and therefore our data, and those from other authors (28), offer no support to the theory that high insulin concentrations depress lipolysis and enhance lipogenesis in extramammary tissues at the expense of milk lipid production.

Phenylalanine, Lys, and Met were the EAA with the most stable arterial concentrations among treatments (coefficients of variation among treatment means were 4, 6, and 8%, respectively), which further supports the theory that in the current rations, Lys and Met were among the most limiting for milk protein synthesis. In general, arterial concentrations of AA were higher with the 18% CP rations than with the 15% CP rations. This difference tended to be significant with the branched-chain AA and was clearly significant in the case of Gly ($P < 0.01$). Arterial Tau concentrations were greater ($P = 0.07$) with the H than the L rations, which could be a consequence of the greater Met supplied by the H

Table 8. Effects of dietary CP and the AA profile offered to dairy cows in early lactation on mammary uptake (mmol/h) of plasma amino acids.

Metabolite	Dietary treatment ¹				SE	Probability ²		
	18H	18L	15H	15L		AA	CP	CP × AA
Alanine	34.3	17.3	27.4	39.0	10.50	0.80	0.51	0.22
Arginine	31.1	34.6	26.4	28.0	4.33	0.57	0.24	0.83
Asparagine	17.3	13.7	14.8	14.6	5.33	0.74	0.89	0.76
Aspartate	2.7	1.4	2.8	2.4	0.87	0.41	0.56	0.61
Citrulline	9.1	3.2	7.3	2.0	4.60	0.27	0.77	0.95
Cysteine	0.20	0.02	0.23	0.16	0.18	0.48	0.70	0.81
Glycine	35.7	8.1	14.0	11.0	15.09	0.35	0.56	0.45
Histidine	12.7	9.3	12.3	10.3	2.89	0.40	0.61	0.81
Isoleucine	43.6	49.0	36.2	32.3	11.42	0.95	0.33	0.70
Leucine	50.7	55.6	44.5	47.4	6.86	0.59	0.33	0.88
Lysine	41.9	41.5	37.8	38.8	6.99	0.97	0.64	0.92
Methionine	9.4	8.9	11.2	9.9	1.71	0.64	0.45	0.82
Ornithine	14.8	10.7	14.5	13.7	2.20	0.31	0.57	0.48
Phenylalanine	18.7	18.0	16.5	16.4	3.63	0.88	0.65	0.98
Proline	8.5	11.0	13.9	11.5	2.59	0.99	0.30	0.40
Serine	25.4	19.1	23.8	23.6	6.46	0.63	0.83	0.65
Taurine	4.2	3.9	5.1	1.7	2.38	0.46	0.80	0.52
Threonine	27.3	24.4	19.9	23.6	5.56	0.95	0.49	0.58
Tryptophan	5.4	6.0	4.9	5.3	1.44	0.32	0.96	0.77
Tyrosine	17.2	13.3	14.9	15.2	3.56	0.62	0.96	0.57
Valine	41.2	44.5	44.3	36.9	10.30	0.85	0.84	0.62

¹18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

²Probability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: Crude protein level in the ration (18 or 15).

Table 9. Effects of dietary crude protein level and the amino acid profile offered to dairy cows in early lactation on arterial plasma or serum concentrations^a of ammonia, urea, amino acids, nonesterified fatty acids (NEFA), glucose and insulin.

Metabolite	Dietary treatment ^b				SE	Probability ^c		
	18H	18L	15H	15L		AA	CP	CP × AA
Ammonia	142.45	182.71	135.13	202.55	44.31	0.27	0.89	0.77
Urea, mM	6.43	5.87	4.97	3.51	0.25	<.01	<.01	0.13
Alanine	198.51	196.36	197.94	238.20	33.38	0.59	0.56	0.55
Arginine	67.56	66.08	56.19	52.95	7.99	0.78	0.18	0.92
Asparagine	44.22	46.19	38.73	42.86	7.94	0.71	0.60	0.90
Aspartate	10.10	9.94	9.76	10.86	0.72	0.54	0.70	0.42
Citrulline	92.49	98.56	88.43	75.58	8.52	0.70	0.16	0.31
Cysteine	3.03	3.65	2.90	3.13	0.25	0.15	0.24	0.47
Glycine	232.14	275.28	294.11	295.90	18.33	0.27	<.01	0.30
Histidine	41.48	33.92	34.83	29.51	4.36	0.19	0.25	0.81
Isoleucine	101.27	120.12	81.00	95.18	12.45	0.23	0.12	0.86
Leucine	146.93	138.57	116.28	113.74	14.38	0.72	0.10	0.85
Lysine	65.74	67.98	59.14	60.99	7.74	0.80	0.41	0.98
Methionine	18.05	16.48	18.93	16.11	2.48	0.41	0.92	0.81
Ornithine	34.40	30.93	27.45	27.19	3.41	0.60	0.17	0.66
Phenylalanine	41.03	41.34	38.85	37.46	3.85	0.89	0.46	0.83
Proline	58.91	54.68	49.54	57.19	7.90	0.84	0.68	0.48
Serine	61.72	65.05	65.52	68.49	61.72	0.69	0.65	0.98
Taurine	38.93	37.18	39.54	30.81	2.35	0.07	0.27	0.19
Threonine	98.30	105.58	82.47	94.98	12.46	0.46	0.33	0.84
Tryptophan	29.80	31.55	30.06	25.40	3.16	0.66	0.39	0.35
Tyrosine	44.52	43.37	41.08	36.90	4.49	0.58	0.31	0.75
Valine	188.99	194.67	148.42	161.20	18.44	0.63	0.09	0.85
Glucose, mg/dL	66.75	65.99	71.32	66.35	4.87	0.58	0.63	0.68
NEFA, μ eq/L	287.82	269.93	406.72	251.32	48.90	0.13	0.35	0.21
Insuline, μ IU/mL	10.93	9.68	8.42	8.20	0.80	0.38	0.04	0.54

^aConcentrations are expressed in μ M unless otherwise indicated.

^b18H: Diet with 18% CP and a high quality AA profile; 18L: Diet with 18% CP and a low quality AA profile; 15H: Diet with 15% CP and a high quality AA profile; 15L: Diet with 15% CP and a low quality AA profile.

^cProbability corresponding to the hypothesis of no effect of AA, CP, or their interaction. AA: Quality of the AA profile (High or Low); CP: Crude protein level in the ration (18 or 15).

rations. Arterial urea concentrations obtained with the 18% CP rations were greater ($P < 0.01$) than those obtained with the 15% CP rations, which indicates that AA oxidation was greater with the 18 than with the 15% CP rations. Also, the H rations resulted in greater ($P < 0.01$) plasma urea concentrations compared with their L counterparts. This observation suggests that the quality of the AA profile in the rations 18H and 15H could be further improved by decreasing the concentrations of certain AA that apparently were oxidized in greater amounts in the H rations compared with the L rations resulting in a consequent increase in arterial urea concentrations.

CONCLUSIONS

Rations containing 15% CP with a good AA profile offered to cows in early lactation can result in the same milk and milk protein yields as rations containing 18% CP and a poor AA profile, which implies a greater efficiency of nitrogen utilization for milk protein synthesis. In early lactation, the effects of the AA profile of the

dietary protein are greater than those of the amount of protein in the diet. Offering 18% CP rations during early lactation may also result in greater arterial and milk urea concentrations compared with 15% CP rations. In rations similar to the those used in this study, lysine and methionine appear to be the first-limiting AA for milk production in early lactation.

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